

ABSTRACT

Predicting NCAA March Madness Outcomes

The aim of this project is to develop a machine learning framework to predict outcomes for the NCAA Division I Mens Basketball Tournament. The purpose is to build upon traditional statistical rankings by implementing an automated pipeline that generates precise win probabilities for every possible tournament matchup. Extensive research into historical sports forecasting and modern machine learning techniques was conducted to establish a baseline for predictive accuracy and model calibration.

Following a comprehensive review of existing literature, the models were constructed using R. The primary dataset includes historical game results and Massey Ordinals, supplemented by custom feature engineering designed to capture team momentum and efficiency metrics. Four different modeling techniques were evaluated on the training data: Logistic Regression, Linear Regression, Random Forest, and XGBoost.

Unlike standard classification tasks, the final model was evaluated based on its ability to minimize the Brier Score, ensuring that the predicted probabilities were well calibrated against actual tournament results. The findings indicate that the integration of ensemble methods and strength of schedule adjustments provides a competitive edge in forecasting the high variance environment of the tournament. The complete codebase and dataset configurations are hosted on GitHub for further research and reproducibility. (do i put the link here or somewhere else)

Jonathen Marin
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Jonathen Marin

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APPROVED

For the Department of Mathematics:

We, the undersigned, certify that the project of the following student meets the required standards of scholarship, format, and style of the university and the student's graduate degree program for the awarding of the master's degree.

Jonathen Marin

Project Author

Earvin Balderama (Chair)

Mathematics

Patrick Chung

Mathematics

Mario Bañuelos

Mathematics

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INTRODUCTION

The National Collegiate Athletic Association (NCAA) serves as the premier developmental system for the National Basketball Association (NBA) and the Women's National Basketball Association (WNBA). With thousands of athletes across both divisions arriving from around the world, the influence of the NCAA is extensive. As the basketball world has evolved into a multibillion dollar industry [1], the value of a competitive edge has become substantial. Teams historically relied on live scouting and film study to gain these advantages. However, the last decade has seen a major shift toward statistical findings. Inspired largely by the movie *Moneyball*, teams now look to analytics to uncover value that the naked eye might miss.

The focal point of this competitive landscape is March Madness. At the conclusion of the regular season, the selection committee invites 68 teams to participate in this tournament. The field includes the champions from all 32 Division I conferences alongside the best remaining teams chosen at large. The committee seeds these participants into four distinct regions with rankings ranging from 1 to 16. The four teams judged to be the strongest are awarded the number 1 seeds while the next tier receives the number 2 seeds. Within each region, the teams compete in a single elimination format where matchups are initially determined by rank. For example, the top seed plays the sixteenth seed. The winner of each region advances to the Final Four to compete for the national championship. Crucially, these contests take place at neutral sites rather than on home courts. This structure creates a predictive challenge where volatility is the norm.

To evaluate these predictive challenges in a rigorous and competitive environment, this project utilizes the Kaggle March Machine Learning Mania competition as its primary experimental framework. Kaggle provides a unique platform where data scientists from across the globe compete to minimize error in tournament forecasts using historical data stretching back to 1985. Unlike traditional bracket challenges that reward binary picks, this competition requires the submission of win probabilities for every possible matchup in the tournament field. This shift from simple selection to probabilistic estimation allows for a more granular analysis of model performance and calibration. By using this structured setting, the research can leverage standardized datasets including team efficiency ratings and historical results to build a pipeline that is both reproducible and statistically sound.

This statistical consistency is possible because basketball is uniquely suited for machine learning applications. Both individual and collective performances are captured through a standardized set of metrics. Discrete variables such as points, steals, assists, and rebounds provide a consistent numerical foundation for analysis across different eras and levels of competition. This uniformity allows for the creation of structured datasets where team strength can be quantified and compared through objective performance indicators. By leveraging these recurring statistical patterns, machine learning models can identify underlying trends that govern game outcomes and team efficiency.

A basic assessment often relies on a record of wins and losses to infer the overall quality of a team. While these records serve as strong indicators,

they frequently fail to capture critical nuance. Furthermore, the single elimination structure of the tournament ensures that upsets remain a constant possibility regardless of historical dominance. A defining example occurred in when the top ranked Virginia Cavaliers entered the postseason with 31 wins and only 3 losses. They fell to the lowest ranked UMBC Retrievers by 20 points. Such outcomes demonstrate why simple heuristics fail in this environment.

This inherent volatility drives a massive secondary market focused on sports betting. As legal wagering expands globally, March Madness has evolved into one of the largest gambling events in the world with 3.1 billion dollars wagered in 2025 on tournament outcomes [2]. To manage this risk, bookmakers generate odds which serve as the industry standard for predictive accuracy. These odds represent a market consensus that attempts to price in every available variable from player injuries to historical trends. However, these lines are often influenced by public sentiment and profit margins rather than pure probability. Consequently, developing a model capable of outperforming these market established baselines represents the gold standard for predictive success.

This paper aims to construct a predictive framework capable of meeting this standard. The primary objective is to develop and evaluate distinct statistical models that minimize the Brier Score metric used in the Kaggle competition. By testing various algorithms against historical data, the research seeks to identify a strategy that generates a winning score. The final analysis benchmarks these models against competitor submissions that utilized multiple strategies including elo ratings, standard odds conversion,

and analysis rankings. Ultimately, the optimal model will be deployed in the upcoming 2026 March Madness tournament to compare its projections with live results.

Motivated by this challenge, substantial research has addressed the complexities of March Madness prediction. The majority of this work focuses on engineering feature sets for learning algorithms ranging from linear regression to more complex ensemble methods. While the Kaggle competition provides foundational game statistics, transforming these into effective predictors requires significant effort. The following subsection details the specific data types provided by the competition to establish this baseline. The final subsection we review existing literature with a primary focus on feature engineering. Finally, the paper details the implementation of distinct modeling techniques. This comparative analysis examines linear and logistic regression alongside additional statistical methods. These models are compared against one another and evaluated relative to the winning scores from prior competitions.

Data Available

The highly quantifiable nature of basketball ensures that statistical data is both abundant and accessible. While the Kaggle competition permits participants to incorporate any external datasets they acquire, the organizers themselves provide a comprehensive archive. This repository includes over 30 distinct tables of historical data spanning more than three decades. All of the tables are separated by gender, with the only exception being the public rankings table. This is due to the fact that they only compiled mens rankings.

The data provided by Kaggle is split into regular season and postseason files. For the purposes of this project, the tournament phase is referred to as the playoffs. This research focuses primarily on the detailed results tables for both the regular season and the playoffs while also utilizing the essential rankings and seeding tables. Although the broader archive extends further back, the detailed game statistics required for this analysis commence in the 2003 season. Each entry in these tables provides a comprehensive statistical breakdown for both the winning and losing teams in every recorded matchup.

Season	DayNum	WTeamID	WScore	LTeamID	LScore	WLoc	NumOT	WFGM	WFGA	WFGM3	WFGA3	WFTM	WFTA	WDR	WDR	WAsst	WTO	WSR	WBlk	WPF	LFGM	LFGA	LFGM3	LFGA3	LFTM	LFTA	LOR	LDR
2003	10	1104	68	1328	62	N	0	27	58	3	14	11	18	14	24	13	23	7	1	22	22	53	2	10	16	22	10	22
2003	10	1272	79	1393	63	N	0	26	62	8	20	10	19	15	28	16	13	4	4	18	24	67	6	24	9	20	20	25
2003	11	1296	73	1437	61	N	0	24	58	8	18	17	29	17	26	15	10	5	2	25	22	73	3	26	14	23	31	22
2003	11	1296	56	1457	50	N	0	18	38	3	9	17	31	6	19	11	12	14	2	18	18	49	6	22	8	15	17	20
2003	11	1409	77	1208	71	N	0	30	61	6	14	11	13	17	22	12	14	4	4	20	24	62	6	16	17	27	21	15
2003	11	1458	81	1186	55	H	0	26	57	6	12	23	27	12	24	12	9	9	3	18	20	46	3	11	12	17	6	22
2003	12	1161	89	1236	62	H	0	23	55	2	8	32	39	13	18	14	17	11	1	25	19	41	4	15	20	28	9	21
2003	12	1186	75	1457	61	N	0	28	62	4	14	15	21	13	35	19	19	7	2	21	20	59	4	17	17	23	8	25
2003	12	1194	71	1156	66	N	0	28	58	5	11	10	18	9	22	9	17	9	2	23	24	52	6	18	12	27	13	26
2003	12	1458	84	1296	56	H	0	32	67	5	17	15	19	14	22	11	6	12	0	13	23	52	3	14	7	12	9	23
2003	13	1166	106	1426	59	H	0	41	69	15	25	9	13	15	29	21	11	10	6	16	17	52	4	11	12	17	8	15
2003	13	1202	74	1106	73	N	0	29	51	7	13	9	11	6	21	18	15	7	1	5	29	63	10	22	5	5	13	16
2003	13	1237	66	1135	65	N	0	26	66	5	19	9	13	21	23	15	17	12	3	17	24	56	6	19	11	17	14	21
2003	13	1323	76	1125	48	H	0	25	56	10	23	16	23	8	35	18	13	14	19	13	18	64	8	24	4	8	14	26
2003	14	1125	83	1135	77	N	1	30	70	11	31	12	15	15	18	22	16	18	5	21	28	69	4	15	17	27	20	27
2003	14	1156	78	1236	71	N	0	27	46	10	18	14	24	8	25	18	15	2	6	18	23	50	7	20	18	24	6	17
2003	14	1161	81	1194	56	H	0	22	48	5	12	32	43	16	32	13	24	5	3	19	21	65	2	13	12	19	16	18
2003	14	1186	82	1202	57	H	0	33	61	8	19	8	14	9	23	21	17	14	5	18	23	55	2	11	9	14	13	23
2003	14	1183	73	1129	59	A	0	29	59	9	19	6	8	8	21	17	15	10	1	19	21	51	3	14	14	18	12	22
2003	14	1314	85	1336	55	H	0	32	60	5	15	16	23	12	34	18	14	8	4	12	21	71	3	23	10	11	15	20
2003	14	1323	89	1237	45	H	0	34	62	8	16	13	21	12	34	25	11	9	13	16	18	74	4	15	5	11	27	22
2003	14	1353	69	1162	36	H	0	23	57	4	19	10	15	15	18	10	12	14	6	18	12	39	3	8	9	13	11	23
2003	14	1390	61	1131	57	H	0	20	53	6	27	15	24	15	24	9	14	4	2	17	19	54	3	17	16	18	11	22
2003	14	1426	59	1106	47	N	0	25	53	1	5	8	12	15	32	14	16	6	4	7	20	60	6	18	1	2	8	15
2003	14	1462	87	1389	48	H	0	35	70	6	19	11	16	17	27	18	7	13	3	10	20	63	7	20	1	4	16	18
2003	15	1161	77	1156	63	H	0	27	58	5	10	18	27	18	26	15	12	5	4	21	23	51	3	13	14	20	7	19
2003	15	1194	93	1236	86	N	2	33	75	9	20	18	33	17	26	14	14	17	0	21	27	58	14	31	18	22	9	33
2003	15	1196	76	1256	55	H	0	28	48	4	13	16	24	8	23	18	18	13	5	19	19	49	1	5	16	22	12	17
2003	15	1242	81	1221	57	H	0	28	53	3	5	22	37	13	29	12	19	10	3	21	18	52	7	21	14	21	8	22
2003	15	1422	84	1447	65	H	0	33	53	3	8	15	20	7	36	12	22	5	9	19	26	70	3	25	10	16	14	15
2003	16	1314	71	1353	67	H	0	27	57	5	14	12	19	14	30	19	20	8	5	13	27	66	9	22	4	11	12	19
2003	16	1390	63	1462	62	H	0	23	57	6	15	11	21	17	20	12	10	8	1	19	18	45	6	20	20	28	11	17
2003	17	1196	99	1183	65	H	0	39	73	11	28	10	22	16	31	32	14	10	14	19	22	66	4	11	17	24	15	23
2003	17	1304	68	1147	45	N	0	27	58	6	18	8	13	10	28	13	17	5	5	26	10	50	2	16	23	31	16	24
2003	18	1104	82	1186	56	H	0	24	48	10	20	24	34	12	26	13	13	7	3	14	19	55	8	21	10	12	11	18

Figure 1. Regular Season Detailed Data

This table records key performance metrics for each game played. The schema utilizes the prefix W to denote statistics for the winning team and the prefix L for the losing team. The specific recorded variables include:

- Points Scored
- The number of field goals made and attempted (FGM-A).
- The number of 3-point field goals made and attempted (FGM-A3).

- The number of free throws made and attempted (FTM-A).
- The number of offensive and defensive rebounds secured (OR- DR).
- The number of assists (Ast).
- The number of turnovers (TO).
- The number of blocks (Blk).
- The number of fouls (PF).

While raw box scores provide the foundation of events that happened in a game, they often fail to capture nuances in, and out, of the game. To address this limitation, this research supplements the Kaggle datasets with advanced metrics from analyst Ken Pomeroy (KenPom). Pomeroy is accepted as the industry standard for college basketball analysis. Pomeroy's approach is to normalize the data by calculating efficiency rather than raw stats. These statistics are updated daily throughout the regular season and tournament. For the purpose of this study, a snapshot of the rankings was acquired from the day prior to the start of the 2025 tournament.

2026 Pomeroy College Basketball Ratings													
help													
02-03-04-05-06-07-08-09-10-11-12-13-14-15-16-17-18-19-20-21-22-23-24-25-26													
Data through games of Monday, January 19 (3689 games)													
Rk	Team	Conf	W-L	NetRtg	ORTg	DRtg	AdjT	Luck	Strength of Schedule			NCSOS	
									NetRtg	ORTg	DRtg	NetRtg	
1	Michigan	B10	16-1	+36.00	127.0	91.0	73.5	+0.22	144	+11.23	116.6	105.4	+11.45
2	Arizona	B12	18-0	+34.04	126.9	92.9	71.8	+0.33	122	+6.59	114.8	108.2	+2.69
3	Duke	ACC	17-1	+32.78	126.0	93.2	67.6	+0.78	43	+9.69	115.7	106.0	+6.04
4	Purdue	B10	17-1	+31.31	129.5	98.2	65.9	+0.87	35	+9.41	116.1	106.7	+7.23
5	Houston	B12	17-1	+31.14	124.1	93.0	63.6	+0.54	78	+5.44	113.5	108.1	+0.85
6	Gonzaga	WCC	19-1	+31.04	126.3	95.2	70.3	+0.96	23	+5.48	113.1	107.7	+7.66
7	Illinois	B10	15-3	+30.44	129.3	98.9	66.4	+0.04	180	+8.62	115.5	106.9	+3.56
8	Iowa St.	B12	16-2	+30.00	125.0	95.0	68.2	+0.33	121	+5.29	112.3	107.0	-2.15
9	Florida	SEC	13-5	+29.57	125.6	96.1	70.4	-0.90	343	+12.33	117.9	105.6	+6.97
10	Michigan St.	B10	16-2	+29.32	120.0	90.7	66.3	-0.32	256	+8.60	116.7	108.1	+2.69
11	Connecticut	BE	18-1	+28.74	121.9	93.2	65.5	+0.87	34	+9.44	114.6	105.2	+7.69
12	Vanderbilt	SEC	16-2	+28.13	127.1	98.9	71.2	+0.32	123	+7.96	115.1	107.1	+1.72
13	Nebraska	B10	18-0	+27.06	123.5	96.5	67.3	+0.92	29	+5.94	113.8	107.9	-4.60
14	Virginia	ACC	16-2	+26.84	124.7	97.9	65.8	+0.02	185	+4.52	113.2	108.7	-3.64
15	BYU	B12	16-2	+26.70	125.2	98.5	70.8	+0.40	105	+7.19	113.9	106.8	+3.71
16	Louisville	ACC	13-5	+26.05	126.9	100.9	70.3	-0.54	300	+7.76	113.2	105.4	+2.11
17	Alabama	SEC	13-5	+25.46	129.4	104.0	73.8	-0.04	198	+14.83	119.1	104.2	+12.50
18	Iowa	B10	13-5	+25.37	123.0	97.7	63.9	-0.70	321	+4.71	112.3	107.6	-7.12
19	Kansas	B12	13-5	+25.25	122.0	96.8	67.8	-0.16	232	+12.99	116.4	103.4	+10.51
20	St. John's	BE	13-5	+24.44	123.1	98.7	70.9	-0.64	312	+9.28	115.8	106.5	+7.09

Figure 2. The first 20 teams of the KenPom data table.

The primary metrics utilized from this source include:

- Net Rating (NetRtg).
- Offensive Rating (ORTg).
- Defensive Rating (DRtg).
- Adjusted Tempo (AdjT).

Net rating is the differential between Offensive and Defensive Rating.

Offensive Rating measures the points scored per 100 possessions. Similarly, defensive rating measure the points given up per 100 possessions. These ratings are adjusted by the quality of opponents a team has played, and the game location. Lastly, Adjusted tempo is the number of possessions per 40 minutes. We define a possession as the period where a team controls the ball, ending with a shot attempt, turnover, or score.

While Pomeroy’s ratings serve as the industry standard, they are exclusively available for the Men’s division. Consequently, this research incorporates an additional dataset from Bart Torvik that focuses on the Women’s division. Torvik cites Pomeroy as a significant influence; thus, the ratings are constructed using a comparable methodology. However, distinct algorithmic differences exist between the two systems. For further detail regarding these distinctions, the reader is referred to [3]. The metrics used from Barttorvik are:

- Adjusted Offensive Efficiency (ADJOE).
- Adjusted Defensive Efficiency (ADJDE).
- Adjusted Tempo (ADJ T.).

RK	TEAM	CONF	G	REC	D1 AVG.	118.2	ADJOE	ADJDE	BARTRNG	EFF FG% 51.3	EFF 3PT% 17.2	TOURNOVS% 31.6	TOURD	REBOUNDS% 31.6	DSHS	DSHS	FT RATE 35.6	FT RATE 35.6	2PT % 51.7	3PT % 33.8	3P RATE 35.7	ADJ T	WAD		
1	Michigan	B10	17	16-1		128.8	91.0	.9818		59.3	42.3	16.1	17.2	35.0	28.4	40.3	27.4	63.9	40.3	35.4	30.1	42.8	41.7	73.6	+5.6
2	Houston	B12	18	17-1		125.4	92.7	.9701		52.1	45.7	13.2	23.7	38.6	30.9	25.0	38.1	52.8	45.7	34.2	30.5	42.1	43.3	63.6	+4.8
3	Arizona	B12	18	18-0		125.8	93.6	.9676		57.0	45.8	16.2	18.2	40.3	25.0	43.1	29.1	57.9	44.3	36.5	32.3	27.4	36.1	71.8	+6.3
4	Connecticut	BE	19	18-1		122.2	92.4	.9616		55.3	43.6	16.6	19.5	34.6	29.1	31.9	38.8	57.2	44.8	34.7	27.5	37.6	33.8	65.3	+6.4
5	Purdue	B10	18	17-1		129.3	98.1	.9598		58.8	48.8	14.2	16.4	37.0	26.7	28.1	22.1	58.9	51.0	39.2	30.6	38.4	44.5	65.7	+5.7
6	Illinois	B10	18	15-3		130.6	99.3	.9591		55.8	45.9	14.6	12.3	39.7	27.5	33.9	19.1	58.0	45.1	35.1	31.4	48.6	40.6	66.3	+4.0
7	Vanderbilt	SEC	18	16-2		128.6	97.8	.9586		57.8	46.8	13.1	18.7	30.8	27.5	38.1	43.0	59.9	49.1	36.8	28.8	44.3	40.3	70.9	+4.3
8	Virginia	ACC	18	16-2		126.5	97.0	.9546		56.0	43.5	15.8	16.4	40.6	30.4	36.4	35.2	56.3	43.6	37.1	28.9	46.5	37.2	65.6	+3.8
9	Florida	SEC	18	13-5		125.7	96.6	.9542		52.4	47.1	17.0	15.2	43.5	23.2	40.8	36.0	59.0	45.5	28.4	33.6	40.4	33.5	70.0	+2.6
10	Gonzaga	WCC	20	19-1		124.8	96.0	.9536		57.7	45.6	14.5	20.5	37.3	24.5	30.6	32.0	58.8	46.6	36.8	29.6	29.9	44.7	70.3	+5.1

Figure 3. The first 10 teams of the Barttorvik table

To complement the efficiency metrics provided by Pomeroy and Torvik, this research also incorporates the comprehensive ranking systems aggregated by Kenneth Massey. While the previously discussed datasets focus on raw predictive efficiency, the Massey Ordinals provide a consolidated view of team hierarchy. This dataset aggregates rankings from over 40 distinct

mathematical systems, pollsters, and algorithms. These rankings typically present data as an ordinal integer where the top team is assigned the value of 1. Incorporating this data allows the model to leverage a "wisdom of the crowd" approach by utilizing the consensus of the broader analytical community.

TEAM	SORT	DP	EBP	EMK	ENG	ESR	FAS	HAS	INC	JNG	KPK	LOG	MAS	MB	MGS
Arizona	>>	4	5	2	2	1	1	4	2	1	3	2	1	1	2
Michigan	>>	1	2	1	1	6	3	1	1	4	1	1	6	6	4
Duke	>>	2	1	3	7	4	4	2	3	2	2	3	2	5	5
Purdue	>>	7	6	5	3	5	6	3	5	6	4	5	4	4	3
Gonzaga	>>	3	3	9	9	7	7	7	6	9	5	4	5	7	1
Connecticut	>>	13	11	6	5	3	5	15	7	5	6	7	7	3	6
Houston	>>	5	4	4	6	8	9	8	4	7	8	12	3	8	9

Figure 4. A sample of the Massey Ordinals table showing distinct ranking systems.

One significant challenge in aggregating these distinct sources is the inconsistency in team nomenclature. For instance, a university might be listed as "St. Mary's" in one dataset and "Saint Mary's CA" in another. To resolve this discrepancy, all external data was strictly mapped to the unique TeamID integer provided by the Kaggle competition files. This standardization ensures that the advanced efficiency metrics are accurately merged with the box score data for every specific matchup.

LITERATURE REVIEW

Past Strategies

The pursuit of a perfect bracket has generated a substantial body of research in predictive modeling over the last decade. While specific algorithms vary, the most successful strategies in the Kaggle Machine Learning Mania competition consistently rely on one of two distinct methodological frameworks.

The primary objective for this study is to minimize the Brier Score. We define the Brier Score (BS) as follows:

$$(1) \quad BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2$$

where N is the number of events, f_t is the predicted probability, and o_t is the actual outcome. For this research, a 0 is considered a loss, and a 1 is a win. A Brier Score of 0 would reflect perfect accuracy, while a Brier Score of 1 reflects perfect inaccuracy. To provide a baseline, a strategy where we predict every team has a 50% chance of winning would result in a Brier Score of 0.25. (thinking of putting another example here)

The first framework relies on fundamental analysis where the objective is to construct a ground-truth probability independent of public sentiment. A common implementation of this approach involves utilizing domain-specific heuristics such as the elo rating system. Originally designed for chess, elo provides a dynamic measure of team strength that updates after every game. It offers a robust baseline that accounts for strength of schedule more

effectively than raw win-loss records [10]. Many strategies also incorporate expert features. These include the Nate Silver, FiveThirtyEight, and KenPom ratings which act as high-signal inputs that aggregate player health, travel distance, and roster depth into a single numeric value. By treating these expert outputs as features rather than final predictions, modelers can leverage human domain expertise while correcting for known expert biases.

The second dominant framework relies on the Efficient Market Hypothesis [8]. This approach posits that global betting markets, specifically closing lines from major bookmakers, offer a more accurate aggregate predictor of game outcomes than individual statistical models. The efficacy of this strategy is well-documented in recent Kaggle competitions where five of the top eight performing models utilized market-derived features as their primary feature. These strategies do not attempt to beat the market. Instead, they focus on mathematically disentangling the true win probability from the bookmaker’s overround and inherent biases.

To achieve this, recent state-of-the-art implementations utilize specific conversion algorithms to de-noise betting data. The standard tool in this domain is the *Goto_conversion*. Competitors employ these functions to address the favorite-longshot bias. This is the tendency for markets to overvalue underdogs and undervalue favorites. By applying these corrections, modelers can transform raw Vegas odds into highly calibrated probabilities that minimize Brier Score more effectively than fundamental analysis alone. This approach effectively leverages the crowd wisdom of the market while stripping away the pricing inefficiencies introduced by public betting behavior. For more information on how this strategy works, we refer the reader to [9].

The applicability of these methodological frameworks, however, is heavily constrained by the structural discontinuity introduced by the 2021 Name, Image, and Likeness (NIL) legislation. Statistical analysis of the post-NIL landscape reveals a significant deviation from historical norms, characterized by a rapid consolidation of talent where the number of programs with elite win percentages ($> 75\%$) has nearly tripled [6]. This phenomenon implies that feature weights derived from pre-2021 eras are likely subject to significant concept drift. Consequently, this literature review restricts its focus to winning strategies from recent Kaggle competitions (2022-present), as methodologies developed before the NIL era may not reflect the approaches best suited to the heightened stratification of team strength driven by current transfer portal dynamics.

Modeling Techniques

Recent tournament prediction methodologies employ several established modeling and computational techniques. This section provides formal definitions for the five core approaches used in this study: Logistic Regression, Linear Regression, Monte Carlo Simulation, Random Forest, and Neural Networks.

Logistic Regression is a binary classification model that estimates the probability of an outcome by applying the logistic (also known as sigmoid) function to a linear combination of input features. The model is defined as:

$$(2) \quad P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

where Y is the binary outcome, $X = (x_1, x_2, \dots, x_n)$ represents the feature vector, and $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through the maximum likelihood estimation. This function ensures the predicted probabilities are bounded between $[0, 1]$. Logistic regression assumes a linear relationship between the input features and the log-odds of the outcome. This assumption can be restrictive when the true relationships are nonlinear. To address this limitation, common approaches are feature engineering with polynomial or interaction terms (e.g., $x^2, x_i \cdot x_j$) to capture nonlinear patterns.

Linear Regression is a supervised learning method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. Linear Regression is defined as:

$$(3) \quad \hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where $\hat{Y}$ is the predicted continuous outcome, $X = (x_1, x_2, \dots, x_n)$ represents the vector of features, and $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through ordinary least squares (OLS) minimization. For more information on OLS we refer the reader to [11]. Unlike logistic regression, linear regression produces unbounded predictions and does not inherently output probabilities. In sports prediction contexts, linear regression is commonly used to estimate point totals, scoring margins, or other continuous performance metrics that can subsequently be transformed into win probabilities through additional techniques.

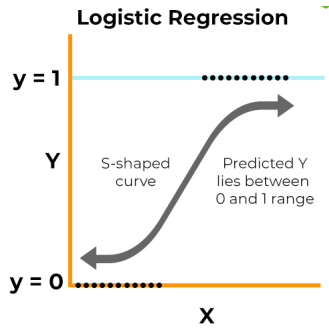


Figure 5. An Example of a Logistic Regression Equation [12].

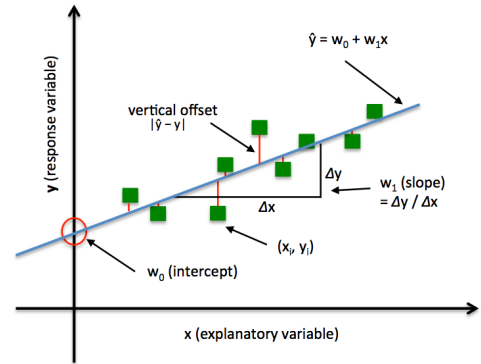


Figure 6. An Example of a Linear Regression Equation [13].

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate probabilities for systems with inherent uncertainty. The method generates N independent samples $x_1, x_2, \dots, x_N$

from a probability distribution $f(x)$, then estimates probabilities by computation the proportion of samples satisfying a certain condition. The simulation is defined as:

$$(4) \quad \hat{P} = \frac{1}{N} \sum_{i=1}^N 1(x_i \in A)$$

where $1(\cdot)$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. A represents the event of interest, and $\hat{P}$ is the Monte Carlo estimate of the true probability $P(A)$. By the Law of Large Numbers [14], $\hat{P} \rightarrow P(A)$ as $N \rightarrow \infty$. In sports prediction contexts, Monte Carlo methods are frequently employed to simulate game outcomes by sampling from estimated score distributions, enabling the estimation of win probabilities through empirical frequency counts across simulated matchups.

XGBoost (Extreme Gradient Boosting), considered a "black-box" approach, is an ensemble learning method that constructs a sequence of decision trees where each successive tree is trained to correct the residual errors of the previous trees. Unlike Random Forest, which builds trees independently, XGBoost employs gradient boosting to iteratively minimize a loss function through additive modeling [5]. The model prediction is defined as the sum of predictions from all trees:

$$(5) \quad \hat{y}_i = \sum_{k=1}^K f_k(x_i).$$

Here, $f_k(x_i)$ is the prediction of the k^{th} tree for the i^{th} data point, $\hat{y}_i$ is the final predicted value for the i^{th} data point, and K is the number of trees.

Each tree is trained to minimize the objective function

$$(6) \quad \text{obj}(\theta) = \sum_i^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where l is the loss function which simply computes the difference between the true and predicted, and Ω is the regularization term, discouraging overly complex trees. The optimization process then proceeds via gradient descent on the residuals. For more information regarding gradient descent we refer the reader to [4]. XGBoost is widely adopted in competitive machine learning for its performance on structured data and ability to capture complex interactions and relationships between features.

$$(7) \quad h^{(\ell)} = \sigma(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$$

if i do neural nets i will keep going with this.

Probability Prediction Models

This section details the development and evaluation of models designed to predict a continuous win probability $P \in (0, 1)$ for any given matchup. While the tournament results are ultimately binary, the Brier Score evaluation metric heavily penalizes overconfidence in incorrect predictions and rewards well-calibrated probabilities. Consequently, this research moves beyond simple classification to focus on models that quantify the uncertainty inherent in the March Madness environment. By establishing a framework where a value $P > 0.5$ denotes a predicted victory for Team A, a baseline is created for performance that accounts for the high variance of single-elimination play. This probabilistic approach serves as the engine of the project and bridges the gap between theoretical literature and the empirical application of machine learning.

To expand further on these concepts, the methodology begins with a rigorous data processing phase where historical box scores are transformed into a symmetric training set. This ensures that the models learn to evaluate the relative strength between two competitors rather than relying on arbitrary formatting artifacts. This section details the complete machine learning pipeline implemented for the 2026 tournament prediction. We first examine the data cleaning procedures required to tailor the feature sets to the specific statistical assumptions of each algorithm. Following this, we explore the variable selection process used to identify high signal metrics such as KenPom and Massey data. Finally, we implement and compare two distinct modeling architectures, specifically Logistic Regression and XGBoost, to determine

which approach best minimizes the Brier Score in a high variance environment.

Logistic Regression

Variable Selection

To accurately model basketball outcomes, one must first deconstruct the sport into its atomic statistical events. A standard game is composed of a finite sequence of possessions. Within these possessions, specific discrete actions determine the flow of the contest. The primary events recorded in the raw dataset include Field Goals, Free Throws, Rebounds, Turnovers, and Personal Fouls. A Field Goal (*FGM*) represents the primary method of scoring and is worth either two or three points. A Free Throw (*FTM*) is an unopposed attempt worth one point awarded after a foul. A Rebound is recorded when a player retrieves the ball after a missed shot; this is further categorized into Offensive Rebounds (*OR*), which grant the shooting team a second opportunity, and Defensive Rebounds (*DR*), which terminate the opponent's possession. Finally, a Turnover (*TO*) is a loss of possession due to a violation or steal, resulting in zero points.

Analyzing these events using raw totals is statistically flawed due to the confounding variable of pace. A team that plays at a high tempo may record high totals for points and rebounds simply because they generate more possessions, not because they are efficient. To resolve this, this model utilizes the "Four Factors of Basketball Success." Developed by Dean Oliver, this framework isolates the four specific components that correlate most strongly with winning: shooting efficiency, turnover avoidance, rebounding dominance, and free throw frequency [16].

Crucial to these efficiency calculations is the estimation of total Possessions ($Poss$). A possession is defined as the period during which a team controls the ball, ending with a shot attempt, a turnover, or a trip to the free-throw line. Since possessions are not explicitly tracked in standard box scores, they must be estimated using the following formula:

$$(8) \quad Poss = (FGA - OR) + TO + (0.44 \cdot FTA).$$

This equation aggregates the three ways a possession can end: a field goal attempt (FGA), a turnover (TO), or free throws (FTA). The term $(FGA - OR)$ accounts for missed shots that are rebounded by the offense; because the team retains the ball, the possession continues, and thus the initial attempt is subtracted to avoid double-counting. The coefficient 0.44 applied to Free Throw Attempts estimates the proportion of trips to the line that actually terminate a possession (e.g., the second of two shots), accounting for and-one (fouled while successfully scoring) situations and technical fouls. This derived metric serves as the denominator for rate statistics, allowing for a direct comparison between teams of differing tempos.

The first factor is Effective Field Goal Percentage ($eFG\%$). This metric refines standard field goal percentage by accounting for the fact that a three-point shot is worth 50% more than a two-point shot. It is calculated as:

$$(9) \quad eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA}$$

By including the 0.5 weighting for three-pointers ($FGM3$), this variable measures a team's ability to maximize points per attempt regardless of their shot selection strategy. The second factor is Turnover Percentage ($TOV\%$). Unlike raw turnover counts, this metric normalizes errors against the total number of possessions calculated above.

$$(10) \quad TOV\% = \frac{TO}{Poss}$$

This ratio identifies teams that value the basketball. A lower percentage indicates a disciplined offense that rarely forfeits scoring opportunities. The third factor is Offensive Rebound Percentage ($ORB\%$). This metric measures a team's physical dominance relative to their available opportunities. Using raw offensive rebound totals is misleading because a team that misses many shots will naturally have more chances to rebound. This metric corrects for that bias by comparing offensive rebounds (OR) to the total number of rebounds available at that specific rim:

$$(11) \quad ORB\% = \frac{OR}{OR + DR_{opp}}$$

Here, DR_{opp} represents the defensive rebounds secured by the opponent. A higher percentage indicates a team that frequently generates "second chance" points. Finally, the model incorporates Win Percentage ($Win\%$) as a fifth feature to capture intangible qualities such as coaching and clutch performance. This is calculated as the mean of the binary game results (1 for a win, 0 for a loss) over the course of the season. Crucially, the logistic

regression algorithm does not process these values in isolation. Instead, the dataset is transformed to calculate the differential between the two competing teams for every metric (e.g., $eFG\%_{diff} = eFG\%_{TeamA} - eFG\%_{TeamB}$). This centers the analysis on the relative mismatch between the opponents rather than their absolute statistics.

Data Cleaning

To construct a robust predictive pipeline, the data architecture must establish a strict boundary between feature generation and model training. The independent variables, specifically the Four Factors and Win Percentage, are calculated exclusively utilizing historical regular-season detailed results. Conversely, the logistic regression algorithm is trained entirely on historical NCAA tournament outcomes. By isolating regular-season efficiency metrics as the predictors and tournament game results as the target variable, the framework ensures that the model evaluates how a team's sustained performance over a thirty-game span translates to the high-variance, single-elimination environment of postseason play. This distinction is critical, as regular-season data provides a sufficiently large sample size to stabilize efficiency metrics, while tournament data provides the specific contextual outcomes the model ultimately aims to predict.

Furthermore, to prevent data leakage and simulate the exact informational constraints of a real-world forecaster, the logistic regression model is trained using a strictly defined rolling time window. To evaluate historical accuracy and construct valid out-of-sample predictions, the framework is systematically back tested across consecutive tournament years.

For example, to generate probability forecasts for the 2025 NCAA Tournament, the model is trained exclusively on the tournament matchups from the preceding 2023 and 2024 seasons. The decision to restrict the training data to a brief temporal window is grounded in the structural realities of collegiate athletics. NCAA basketball is characterized by high roster turnover, as student-athletes possess limited eligibility and typically serve as primary contributors for only three to four seasons. Consequently, a university program undergoes a complete personnel and strategic transformation within a short cycle. Training the algorithm on data exceeding this window would introduce noise from past iterations of a team that share no roster overlap with the current squad. During this localized training phase, the algorithm maps the recent regular-season metrics to their respective tournament results, learning the optimal coefficient weights without accessing any data from the target prediction year. This rolling training window shifts iteratively for historical benchmark predictions, ensuring that all evaluations remain mathematically rigorous and entirely devoid of look-ahead bias.

The primary obstacle in processing the historical game records is the fundamental structure of the Kaggle dataset. The raw regular-season and tournament data files utilize a deterministic outcome schema. Within this structure, every game observation is explicitly bifurcated into winning and losing statistics. Metrics belonging to the victorious program are assigned column headers prefixed with a 'W', while the defeated opponent's metrics are prefixed with an 'L'.

If a classification algorithm is trained directly on this format, it inevitably suffers from severe target leakage. The logistic regression algorithm

would trivially learn to associate the column position with the outcome, mathematically recognizing that the entity occupying the 'W' feature space possesses a 100 percent win probability. This renders the model entirely incapable of out-of-sample prediction, as the ultimate objective is to forecast matchups between two neutral entities before the outcome is known. The following output illustrates this problematic data architecture, demonstrating how the game result is inherently encoded into the feature presentation itself.

```
head(mens_results %>%
      select(Season, WTeamID, WScore,
             LTeamID, LScore, WFGM,
             LFGM, WTO, LTO), 4)
```

##	Season	WTeamID	WScore	LTeamID	LScore	WFGM	LFGM	WTO	LTO
##	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
## 1:	2003	1104	68	1328	62	27	22	23	18
## 2:	2003	1272	70	1393	63	26	24	13	12
## 3:	2003	1266	73	1437	61	24	22	10	12
## 4:	2003	1296	56	1457	50	18	18	12	19

Table 1. Snapshot of raw Kaggle regular-season data demonstrating the 'W' and 'L' column structure.

To resolve the target leakage inherent in the raw dataset, the first phase of data processing transforms the game-level observations into season-level team averages. This transformation is executed via the `calculate_four_factors` function. Initially, the function estimates the total possessions for both the winning and losing teams using the standard formula. Subsequently, it computes the Four Factors (Effective Field Goal Percentage, Turnover Percentage, Offensive Rebound Percentage, and Free Throw Rate) for each team in every game.

Rather than maintaining the winner and loser statistics in a single row, the function separates the victors and the defeated into two distinct dataframes. A binary `Win` variable is appended, assigning a value of 1 to the winning dataframe and 0 to the losing dataframe. These two structures are then vertically concatenated using a row-bind operation. Finally, the data is grouped by `Season` and `TeamID` to calculate the mean of all numeric variables. This yields a single, stable efficiency profile and an aggregate regular season win percentage for every program in a given year.

```
prepare_model_data <- function(tourney_results_df,
                                four_factors_df) {

  training_data <- tourney_results_df %>%
    mutate(
      Team1 = pmin(WTeamID, LTeamID),
      Team2 = pmax(WTeamID, LTeamID),
      Team1_win = if_else(Team1 == WTeamID, 1, 0)
    ) %>%
    select(Season, Team1, Team2, Team1_win)

  model_data <- training_data %>%
    left_join(four_factors_df,
              by = c("Season", "Team1" = "TeamID")) %>%
    rename_with(~paste0(., "_T1"),
               .cols = -c(Season, Team1, Team2, Team1_win)) %>%
```

```

left_join(four_factors_df,
          by = c("Season", "Team2" = "TeamID")) %>%
rename_with(~paste0(., "_T2"),
.col = -c(Season, Team1, Team2, Team1_win, ends_with("_T1"))) %>%
mutate(
  eFG_diff      = eFG_T1 - eFG_T2,
  TOV_Pct_diff  = TOV_Pct_T1 - TOV_Pct_T2,
  ORB_Pct_diff  = ORB_Pct_T1 - ORB_Pct_T2,
  FTR_diff      = FTR_T1 - FTR_T2,
  Win_diff      = Win_T1 - Win_T2
) %>%
na.omit()

return(model_data)
}

```

Model Implementation

Following the systematic cleaning and neutralization of the dataset, the analysis proceeds to the implementation of the predictive framework. To evaluate the impact of regular-season efficiency on postseason success, a series of logistic regression models are constructed. This method is chosen specifically for its ability to model the probability of a binary outcome—in this case, whether the lower-ID team (Team 1) wins or loses—based on the linear combination of the statistical differentials. The models utilize a logit

link function to map the relationship between the independent variables and the probability of a victory. The formal specification for each model iteration is defined by the following equation:

$$(12) \quad \text{logit}(P) = \ln \left(\frac{P}{1-P} \right) = \beta_0 + \beta_1(\text{eFG_diff}) + \beta_2(\text{TOV_diff}) \\ + \beta_3(\text{ORB_diff}) + \beta_4(\text{FTR_diff}) + \beta_5(\text{Win_diff})$$

Where P represents the probability of a Team 1 victory. The coefficients (β) represent the change in log-odds of winning for every one unit increase in the respective statistical differential.

Logistic Regression

Boosting

Results

Here are my insights...

Point Spread Models

same as above

S

Here are my insights...

Comparison of models

My Contributions and Analysis

Here are my insights...

2026 Competition

Strategies

Here are my insights...

Results

CONCLUSIONS

Summary

As a preliminary assessment...

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